

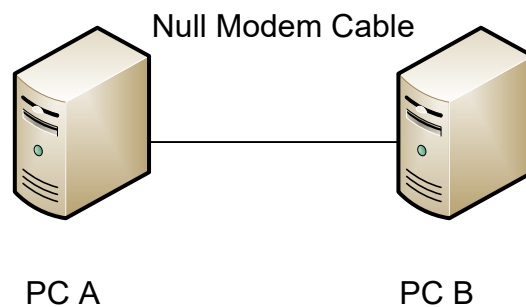
Lab 2

Serial Communications

Objective:

The purpose of this lab is for the student to create a basic network connection between two devices using their serial ports. Students will explore various serial settings using the HyperTerminal program for observation.

Diagram:



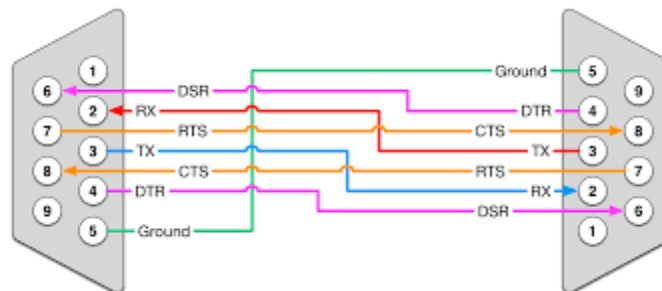
Components:

- 2 – PC's with serial ports
- 1 – Null Modem Serial Cable

Procedure:

1. To make a basic connection between the PC's

- a. Connect the null modem cable to each PC's Com Port
 - b. Start HyperTerminal program. Should be on Desktop. If not, get USB drive from instructor.
 - c. Enter "513" for the area code. Click OK
 - d. Click OK for location.
 - e. Enter a name for the connection. Use "Lab2"
 - f. Select **Connect using Com1**
 - g. Set the **bits per second** to be **9600**
 - h. Set the **data bits** to be **8**
 - i. Set the **parity** to be **none**
 - j. Set the **stop bits** to **1**
 - k. Set **flow control** to be **none**
 - l. Repeat for second PC
2. Type a message on one PC and view it on the other
- a. Does the message appear on both machines or just one?
_____Just one_____
 - b. Why? Think about our types of communication on slide 6.
_____The communication is a half duplex, so it travels one way at a time, but not at the same time _____
3. Draw the pinout connections of the null modem serial cable (pin 1 to pin 1).
Use Drawing Tools in Word or Copy Paste from Google.



4. Make PCA a control PC (settings that do not change) and the other a test PC. Make a connection between the PC's and observe what happens when settings do not match. (Control PC is set to 9600 baud and test PC is 2400 or control PC is 8 data bits and test PC is 5). Record your observations below. You don't have to include the exact characters. Explain what you see. (Crazy characters, undesired results, etc.).

Setting That's Different**Effect**

Bits per Second

_____The characters are completely different; it's gibberish – there's data loss also _____

Data Bits

_____The characters changed to some other kind of symbols. On the test PC, the message from the control included new line characters and some data loss. _____

Parity

_____The characters, once again, are just gibberish, there is no indication of what the original data looked like or what it was supposed to be. _____

Stop Bits

_____The characters are not clear – although it was interesting to see what the difference one -this time the test pc's characters were represented by multiple characters on the control as opposed to just being one to one. _____

Flow Control

_____There was no change in the text; it was clear again _____

5. Put all settings back to the original settings on both PC's. Find a file that is around 10MB in size. Send the file from PCA to PCB and record the time it took to send the file over. Send the file again with baud settings of 2400, 4800, and 9600. Record your times below. Don't wait for the transfer to finish. Wait 10-15 seconds for the time to stabilize, record your time and then cancel. To get your 10 MB file, click start, type "snipping tool", click the tool to launch. Click New and then use the tool to screen shot your desktop. Click Copy, open MS Word and then paste your screen shot several times (15 or so). Click File, Save As, click file type and change to .doc. You don't want it to say docx. Check the size of the file. If you need to adjust, either delete or add additional screen shots to the file.

2400 _____ 10 hours 15 minutes _____

4800 _____ 5 hours 20 minutes _____

9600 _____ 2 hours 43 minutes _____

6. Figure out the time needed to transfer a 2.4Gigabyte file with the following network speeds (remember network speeds are in bits and the file is in bytes). You can use an online file transfer calculator using Google if you prefer.

a. 9.6kb _____ 26 days, 16 hr, 47 min, 47 sec _____

b. 56kb _____ 4 days, 13 hr, 51 min, 2 sec _____

c. 10Mb _____ 36 min, 2 sec _____

d. 100Mb _____ 3 min, 36 sec _____

e. 1Gb _____ 21 sec _____

f. 10Gb _____ 2 sec _____

7. What did you learn by completing this lab?

_____ I learned about the RS232 Protocol with serial connections, more details about connection speed and how the older systems of connection such as serial connections took a very, very long time. Obviously connections with higher network speeds are able to transfer data more readily, but the difference is staggering – weeks get turned into seconds.

